

Fast Bayesian Inference of Reparameterized Gamma Process With Random Effects

Shirong Zhou , Ancha Xu , Yincui Tang , and Lijuan Shen 

Abstract—In the field of reliability engineering, the gamma process plays an important role in modeling degradation processes. However, extracting lifetime information from product degradation observations has long been suffering from both ineffective modeling techniques and inefficient statistical inference methods. To overcome these challenges, we propose a reparameterized gamma process with random effects in this article. Compared with the classical gamma process, the proposed model has a more intuitive physical interpretation. In addition, statistical inference for the model can be readily done through the variational Bayesian algorithm. Combining with the Gauss–Hermite quadrature and the Laplace approximation, the algorithm yields closed-form variational posteriors for the proposed model. Its superiority over two other inference methods (expectation maximization and Monte Carlo Markov Chain) in terms of computational efficiency and estimation accuracy is demonstrated by simulation.

Index Terms—Degradation analysis, gamma process, Gauss–Hermite (GH) quadrature, Laplace approximation, variational Bayesian (VB) approach.

I. INTRODUCTION

A. Background

EFFECTIVE lifetime assessment is the basic requirement for the development and design of new products or materials and can be further used for the warranty and maintenance decisions of equipment or systems. Lifetime-related degradation information (e.g., operating current, wear, and fatigue cracks) is usually collected to infer the lifetime of the system by using

degradation models. However, in order to fully mine degradation information and accurately fit the degradation path of products, degradation models become more and more complex. On the other hand, benefiting from the rapid development of sensor technology, the capacity of collecting degradation data is increasing. The two aspects challenge the statistical inference of system reliability. In the last two decades, Markov chain Monte Carlo (MCMC) methods have been well developed and widely utilized for complex models, which conduct parameter estimation by posterior samples generating from a collection of Markov chains [1]. However, it is noted that MCMC methods could suffer from three well-known drawbacks [2]: First, despite computing hardware has a permanent improvement, MCMC methods still could spend an extremely long time to reach the stationary distribution for large dataset and complex statistical models, even for moderate-scale observations set. Second, the determinations of the length of convergence and burn-in usually are rude. For ensuring the precision of the estimation, it is a common way to take a very large number of iterations in real applications, which further aggravates the calculation burden. Third, real-time estimation is another challenge to MCMC methods. Thus, due to the increasing capacity of degradation observations and the complexity of degradation models, MCMC methods may encounter limitations in some engineering applications.

Therefore, it is desirable to develop an approach that simultaneously has the satisfactory computational efficiency and estimation accuracy for the complex degradation models. One such is the variational Bayesian (VB) approach that originally was developed for efficiently conducting image classification based on mixed Gaussian models [3]. VB can be viewed as a Bayesian approximate inference approach, which tries to seek a tractable (or proxy) distribution to approximate the complex posterior [4]. The discrepancy between the true posterior and its proxy distribution is commonly measured through Kullback–Leibler (KL) divergence, which is a nonnegative and asymmetric similarity measure of two probability density functions (pdfs) [5]. In general, the smaller the KL divergence, the more accurately the proxy distribution can reflect the features of the true distribution. The KL divergence reaches the minimum of zero when the two pdfs are equal. Under the prespecified precision, the optimal distribution can be viewed as the proxy distribution with minimum KL divergence. In this way, VB turns Bayesian inference into an optimization problem. Due to the utilization of the simpler distribution in VB, under satisfactory calculation accuracy, the computation burden and efficiency of the VB approach are usually superior to the traditional Bayesian

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method, for the intricate model with large or medium-sized observation data. For example, the computational efficiency of VB is much higher than MCMC and can reach high precision for the traditional Wiener degradation model with two random effects [6].

B. Related Works

1) *Degradation Modeling*: For assessing the lifetime information of the product with measurements of degradation, many models have been discussed and developed in the literature, such as the general regression-type degradation model [7], stochastic process model [8], and machine learning based model [9], [10]. In contrast, stochastic process degradation models can reflect the time-dependent structure of the deterioration process of the product or component and establish a natural link between the degradation process and the stochastic partial differential equations describing the physical process [11]. Therefore, the stochastic process models are widely used in process monitoring [12], preemptive maintenance [13], and prediction of remaining useful life (RUL) [14]. The Wiener process and its extensions are the most commonly used stochastic process models due to their mathematical properties and physical interpretability. See Whitmore and Schenkelberg [15], Ye et al. [16], Zhai et al. [17], etc. However, for certain types of degradation processes with independent nonnegative increments, the class of Wiener processes may not be appropriate. An alternative option is the gamma process, which can be viewed as the limiting distribution of the compound Poisson distribution when gamma-distributed increments tend to zero and the Poisson rate tends to infinity [18]. An excellent and comprehensive review on gamma-process-based degradation modeling and maintenance was given by van Noortwijk [19]. The basic gamma process $\{Y(t); t \geq 0\}$ with scale parameter $\beta > 0$ and time-varying shape parameter αt ($\alpha > 0$) has independent and gamma distributed increments $\Delta Y(t) = Y(t + \Delta t) - Y(t)$, that is

$$\Delta Y(t) \sim Ga(\alpha \Delta t, \beta)$$

which was first utilized for degradation modeling by Abdel-Hameed [20]. Singpurwalla [21] then gave a comprehensive view of the basic gamma process from the point of methodology and pointed out that it can give a satisfied solution for the deterioration of the skin of aircraft. Park and Padgett [22] developed a closed-form lifetime distribution for the first hitting time of the above basic gamma degradation model and proved that it can be approximated as an inverse Gaussian distribution by considering the discrete damage accumulation. For some materials, there may exist the hidden endogenous variations from themselves and unobservable exogenous diversities caused by their operation environment, which may cause unit-to-unit variability in both the initial values and the degradation path of the samples. Heterogeneity of initial values is generally captured by setting the initial amount of degradation to follow some random distribution, see [23] and [24]. Heterogeneity between degradation paths can be characterized by setting one or more parameters of the model to follow a particular distribution, see [25], [26], [27], and [28]. Some product degradation processes

may be influenced by covariates. For fixed covariates or simple dynamic covariate processes, Lawless and Crowder [29] proposed an extensive gamma degradation based on the relationship between degradation rates and covariates. Performance degradation indicators for products can exhibit linear, convex, or concave overall aging trends. Currently, nonlinear functions are generally used in degradation modeling to convert detection times on a time scale, with exponential and power functions commonly used for conversion. For example, Tseng et al. [30] utilized a gamma degradation model with a power scale function of time to fit the degradation path of a batch of light-emitting diode devices. Similar modeling frameworks can be found in [31] and [32]. In addition, the available degradation observations may be contaminated with noise. Such uncertainty noise can be accommodated by incorporating measurement error into the gamma degradation model. The Wiener process, by its own mathematical nature, assumes that independent measurement errors over time obey a Gaussian distribution with a mean of zero [32]. However, under the assumption of independent Gaussian measurement errors over time, the gamma process leads to a likelihood function that does not have an analytic structure. For this reason, Pulcini [33] proposed a time-dependent first-order autoregressive measurement error to model the measurement error based on the degenerate model of the gamma process in a compromising manner. A similar modeling scheme was proposed by the authors in [34] and [35].

2) *VB Inference*: The VB approach has been shown to be an efficient method in degradation modeling [6], prediction of RUL [2], prognostic and health management [36], etc. There exist two paths to solving engineering's modeling problems with the VB approach in view of methodology. The first is to replace computation-intensive steps of the algorithm with the VB approach in model evaluation. For example, Kraus et al. [9] developed a structured-effect neural network model to forecast the RUL of a turbofan engine, which replaced the structured-effect part of the proportional hazards model with the recurrent neural network and solved the random effects parts of the proportional hazards model with the VB method. The other one directly utilizes the VB approach to conduct model estimation. For example, Zaidan et al. [2] utilized two multilevel Bayesian hierarchical models to predict the RUL of synthetic degradation data and obtained a closed-form solution by using the VB approach. However, the second path may encounter a problem in which the model does not possess conjugated structures for some parameters, which causes the intractable variational posterior. To tackle this challenge, the designed numerical optimization or integration can be applied. For example, Zhou et al. [6] embedded a three-step algorithm in the VB approach for assessing the Wiener degradation model with random effects.

C. Overview

The classical gamma degradation model with random effects has two drawbacks: 1) The random effect is described through the scale parameter, so that the physical interpretation is not direct. 2) It will encounter the problems of the tricky variational

posterior density (VPD) and the unstable numerical integration for VB inference. Thus, a reparameterized gamma (RGa) process and its corresponding VB algorithm are proposed to solve these problems. The main contributions of this article are twofold: 1) An RGa process is proposed, and the parameters therein have evident physical meaning. The random effects can be easily incorporated into the model and diversify the degradation mean and the degradation volatility simultaneously. 2) Combining with the Gauss–Hermite (GH) quadrature [37] and the Laplace approximation [38], a closed-form VB estimator is developed for the RGa degradation model. Then, to tackle the inadequacy of interval estimates of the VB approach, we develop a method for constructing interval estimates based on the observed information matrix evaluated by GH quadrature.

The rest of this article is organized as follows. In Section II, an RGa process is presented for degradation data modeling, and the lifetime distribution of the product is derived. In Section III, a closed-form VB estimator is first developed for the reparameterized gamma degradation model by utilizing the GH quadrature and the Laplace approximation, then the interval estimates are discussed and developed, and at last two benchmark estimators are proposed for comparison. Sections IV and V are devoted to numerical experiments and case study, respectively. Finally, Section VI concludes the article..

II. REPARAMETERIZED GAMMA PROCESS

Let $\{Y(t), t \geq 0\}$ denote the degradation process of a certain performance characteristic of the system. In this study, we assume that $\{Y(t), t \geq 0\}$ follows an RGa process that is characterized by properties: (i) $Y(0) = 0$; (ii) $Y(t_1) - Y(t_2)$ is independent of $Y(t_3) - Y(t_4)$ for any $0 \leq t_i, i = 1, \dots, 4$; (iii) Each increment $\Delta Y(t) = Y(t) - Y(t + \Delta t)$ follows a gamma distribution with shape parameter $\alpha \Delta t$ and scale parameter η/α , denoted by $Ga(\alpha \Delta t, \eta/\alpha)$, $\alpha > 0, \eta > 0$. According to the additivity of gamma distribution, it can be deduced that $Y(t) \sim Ga(\alpha t, \eta/\alpha)$, with pdf

$$p_{Y(t)}(y|\eta, \alpha) = \frac{1}{\Gamma(\alpha t)} \left(\frac{\eta}{\alpha}\right)^{-\alpha t} y^{\alpha t - 1} e^{-\frac{\alpha y}{\eta}}. \quad (1)$$

The respective mean and variance of $Y(t)$ are ηt and $\eta^2 t/\alpha$, and the coefficient of variation is $1/\sqrt{\alpha t}$. Therefore, the parameter η can be viewed as the average degradation rate (or slope) from the view of physics, and the parameter α describes the extent of variability of degradation in a sample in relation to the mean degradation of the population. One main property of the classical gamma process [26] is the dependence on the degradation mean and volatility, which can improve the fitting precision as in the discussion of Ye et al. [39]. It can be easily shown the proposed RGa process does not sacrifice this dependence. Note that the pdf of the proposed RGa process has a similar structure with that of the gamma distribution widely used in the regression. The main difference between them is that the shape parameters of the former vary dynamically with the time of detection while those of the latter are fixed.

In a real degradation test, the observed degradation path may exhibit variability across test units in the same batch, due

to the hidden endogenous variations of the inspected material specimens and unobservable exogenous diversities caused by the operating environments. In order to reflect the unit-to-unit variability in the degradation modeling, it is common to incorporate random effects into the baseline model from a statistical perspective. For the proposed RGa model, it is assumed that its degradation rate parameter η varies from unit to unit and follows a log-normal distribution with location parameter $\mu \in (-\infty, \infty)$ and scale parameter $\sigma > 0$

$$p(\eta|\mu, \sigma^2) = \frac{1}{\eta\sigma\sqrt{2\pi}} e^{-\frac{(\ln(\eta)-\mu)^2}{2\sigma^2}}. \quad (2)$$

The mean and variance of η are $e^{\mu+\sigma^2/2}$ and $(e^{\sigma^2} - 1)e^{2\mu+\sigma^2}$, and the parameters μ and σ^2 can be simply evaluated as $\mu = E_\eta[\ln(\eta)]$, $\sigma^2 = E_\eta[(\ln(\eta) - \mu)^2]$ [40]. The reason and superiorities for modeling the unit-specific heterogeneity of degradation rate η with the log-normal are listed as follows.

- 1) Its domain of definition is positive real numbers, which is suitable for the monotonically increasing and positive degradation quantity. Note that the degradation process of monotonically decreasing can be converted to monotonically increasing by simple mathematical operations.
- 2) It is possible to fit various skewed distributions flexibly. In the final example analysis, we verify that the log-norm distribution can fit well the empirical estimates of the random effects variable η for two sets of real data.
- 3) In the framework of variational inference, an analytic joint variational posterior with respect to the parameters (μ, σ^2) can be constructed with the configuration of log-normal distribution random effects. Thus, in subsequent mean-field variational inference, we can relax the assumption of full factorial decomposition of the mean-field distribution family to be just batch independent. Thereby, the correlation of the parameters (μ, σ^2) is considered in the VB inference. Note that setting η follows other distributions defined over positive real number fields would lead to variational posteriors for the parameters (μ, σ^2) that do not have analytic expressions.
- 4) In the classical gamma process, this unit-to-unit variability is usually incorporated into the scale parameter. Thus, the setup of random effects in the RGa process is more intuitive than that in classical gamma degradation models from the view of physical interpretation.
- 5) The degradation mean for the proposed degradation model only depends on η and is invariant with shape parameter α . For the Bayesian community, it is reasonable to assume that the parameter α is independent of the parameters vector (μ, σ^2) when the priors are assigned, which is critical for the utilization of the mean-field theory in the VB approach.
- 6) In addition, under the fixed inspection frequency Δt , the proposed RGa model with log-normal random effect has a similar structure with the gamma log-normal model [41] that is widely used for modeling the radio channels of the wireless communication systems.

With (1) and (2), the joint pdf of $Y(t)$ and η can be obtained by

$$p_{Y(t),\eta}(y_t, \eta | \Theta) = p_{Y(t)}(y_t | \eta, \alpha) p(\eta | \mu, \sigma^2) \quad (3)$$

where $\Theta = (\alpha, \mu, \sigma^2)$. Then, integrating out η from (3), the marginal pdf of $Y(t)$ is

$$p_{Y(t)}(y_t | \Theta) = \frac{\alpha^{\alpha t} y_t^{\alpha t - 1}}{\Gamma(\alpha t) \sqrt{2\pi} \sigma} \int_0^{+\infty} \eta^{-\alpha t - 1} e^{-\frac{\alpha y_t}{\eta} - \frac{(\ln(\eta) - \mu)^2}{2\sigma^2}} d\eta$$

evaluation of which may lead to a numerically unstable solution, due to the complex integrand and the constrained region of integration. A natural way is by means of reparameterizing or transformation with $z = (\log(\eta) - \mu)/\sqrt{2\sigma^2}$, which leads to

$$p_{Y(t)}(y_t | \Theta) = \frac{\alpha^{\alpha t} y_t^{\alpha t - 1}}{\Gamma(\alpha t) \sqrt{\pi}} \int_{-\infty}^{+\infty} q(z, y_t | \Theta) e^{-z^2} dz \quad (4)$$

where $q(z, y_t | \Theta) = e^{-\alpha t(\mu + z\sqrt{2\sigma^2}) - \alpha y_t e^{-(\mu + z\sqrt{2\sigma^2})}}$.

The integral expression in (4) happens to be the GH quadrature [37]. GH quadrature can be seen as the deterministic version of the Monte Carlo (MC) integration, which utilizes a few predetermined evaluation points to replace the massive random points in MC integration and substitutes special weight functions for the equal weights in MC integration. Thus, the integration of GH quadrature has higher computational efficiency compared with MC integration. More importantly, GH quadrature can reach an approximation in high accuracy like MC integration [42]. With a GH quadrature approach, (4) can be numerically approximated as

$$\int_{-\infty}^{+\infty} q(z, y_t | \Theta) \exp(-z^2) dz \approx \sum_{l=1}^L w(z_l) q(z_l, y_t | \Theta) \quad (5)$$

where $z_l, l = 1, \dots, L$ are the roots of the L th-order Hermite polynomial $H_L(z_l)$ [43], which is formulated as

$$H_L(z_l) = L! \sum_{k=1}^{\lfloor \frac{L}{2} \rfloor} \frac{(-1)^k (2z_l)^{L-2k}}{k!(L-2k)!}$$

and the weight functions $w(z_l), l = 1, \dots, L$ are called GH factors, given by $w(z_l) = (2^{n-1} n! \sqrt{\pi}) (n^2 (H_{L-1}(z_l)))$.

In Fig. 1, we depict the GH-based approximated pdf (5) and the corresponding empirical pdf under four different cases. From Fig. 1, we can see that GH-based approximated pdf (5) performs well for all cases.

Let ω denote the degradation threshold of the tested product. The lifetime T of the product is defined as the first passage time when $Y(t)$ crosses ω , i.e., $T = \inf\{t | Y(t) \geq \omega\}$. Therefore, the cumulative distribution function (cdf) of T is given by

$$F_T(t) = 1 - \Pr(Y(t) \leq \omega) = \int_{\omega}^{+\infty} p_{Y(t)}(y_t | \Theta) dy_t. \quad (6)$$

With the accurate approximation of $p_{Y(t)}(y_t | \Theta)$, (6) can be calculated through the pointwise trapezoidal integral rule. For each inspection point $\tilde{t}$, define $G(y | \tilde{t}, \Theta) = \tilde{p}_{Y(\tilde{t})}(y | \Theta)$, where $\tilde{p}_{Y(\tilde{t})}(y | \Theta)$ is the GH integral approximation of $p_{Y(t)}(y_t | \Theta)$ evaluated at $\tilde{t}$. Then, calculate the integration $\int_{\omega}^{\infty} G(y | \tilde{t}, \Theta) dy$

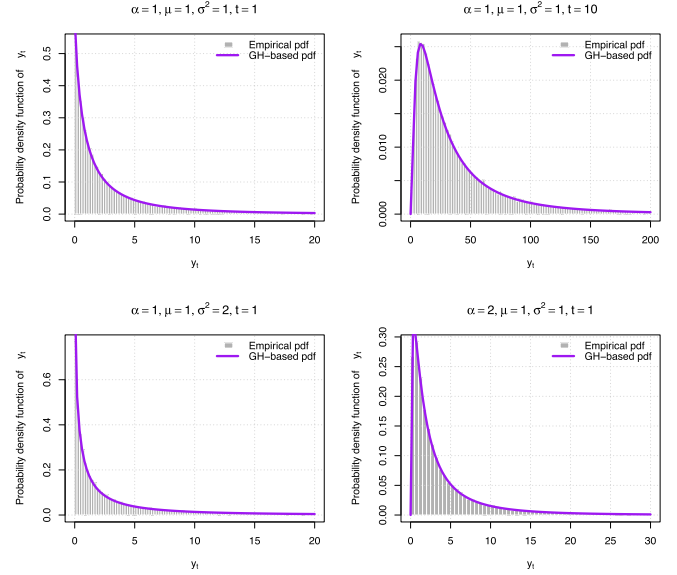


Fig. 1. PDF of y_t under different settings.

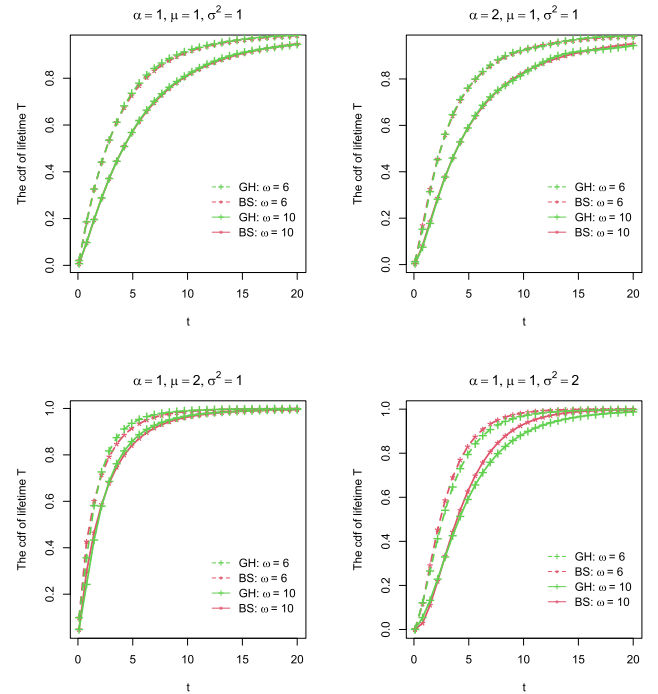


Fig. 2. CDF of lifetime T under different settings.

with the trapz function in pracma [44] R package. This approach matches exactly the approximation for integrating the function using the trapezoidal rule. For validating the overall integration effect, we compare the proposed GH-based integration framework with the Birnbaum–Saunders (BS) approximation of cdf of the lifetime T under different parameters vector Θ and degradation threshold ω in Fig. 2. The results suggest that there is no significant difference between two approaches.

III. STATISTICAL INFERENCE

In a degradation experiment, assume that one special product has a performance characteristic with an increasing degradation path (e.g., laser device [45]). Let $\mathbf{y}_n$ be the observed degradation data for the n th specimen with M_n observations $y_{n,m}$ at each inspection point $t_{n,m}$, $n = 1, \dots, N$, $m = 1, \dots, M_n$, where N represents the total number of all available tested samples. For simplicity, we denote $\mathbf{y} = \{\mathbf{y}_n\}_{n=1}^N$, $\mathbf{y}_n = \{y_{n,m}\}_{m=1}^{M_n}$. Assume that $\mathbf{y}$ are the realizations of the RGA process $\{Y(t), t \geq 0\}$ proposed in the previous section. For symbolic convenience, let $x_{n,m} = \Delta y_{n,m} = y_{n,m} - y_{n,m-1}$, $h_{nm} = \Delta t_{n,m}$, $\mathbf{x}_n = \{x_{n,m}\}_{m=1}^{M_n}$, $\mathbf{x} = \{\mathbf{x}_n\}_{n=1}^N$. In particular, it is assumed that $x_{n,1} = y_{n,1}$, $t_{n,M_n} = t_{n,M}$, $h_{n,m} = h_c$. Then, according to the property (iii) of the RGA process, the increments $\mathbf{x}$ are independent distributed gamma random variables. Given random effects $\boldsymbol{\eta} = \{\eta_n\}_{n=1}^N$, the pdf of $x_{n,m}$ is given by

$$p_{x|\eta_n}(x_{n,m}|\alpha h_c, \eta_n) = \frac{x_{n,m}^{\alpha h_c - 1}}{\Gamma(\alpha h_c)} \left(\frac{\eta_n}{\alpha}\right)^{-\alpha h_c} e^{-\frac{\alpha x_{n,m}}{\eta_n}} \quad (7)$$

and the hidden variable set $\boldsymbol{\eta}$ is modeled as independent and identically log-normal distribution proposed in (2), with pdf

$$p(\eta_n|\mu, \sigma^2) = \frac{1}{\eta_n \sigma \sqrt{2\pi}} e^{-\frac{(\ln(\eta_n) - \mu)^2}{2\sigma^2}}. \quad (8)$$

Let $\boldsymbol{\Theta} = (\alpha, \mu, \sigma^2)$. Thus, the joint posterior of $(\boldsymbol{\eta}, \boldsymbol{\Theta})$ can be written by

$$p(\boldsymbol{\eta}, \boldsymbol{\Theta}|\mathbf{x}) \propto \prod_{n=1}^N \left[p(\eta_n|\mu, \sigma^2) \prod_{m=1}^{M_n} p_{x|\eta}(x_{n,m}|\alpha h_c, \eta_n) \right] \times p(\alpha)p(\mu, \sigma^2) \quad (9)$$

where $p(\alpha)$ and $p(\mu, \sigma^2)$ are the priors of the parameter α and the blocking parameters (μ, σ^2) , respectively. It is noticed that the setup of priors in (9) suggests that 1) μ and σ^2 may depend on each other because there is no sufficient evidence to show the correlation between μ and σ^2 ; 2) α is independent with (μ, σ^2) and vice versa, which comes from the fact discussed in the setup of random effects in above section.

A. VB Estimator

Since the complex structure of the joint posterior (9) makes statistical inference difficult, the VB approach alleviates this difficulty by selecting a set of tractable VPDs $q(\boldsymbol{\Theta}, \boldsymbol{\eta})$ to approximate the true posterior densities $p(\boldsymbol{\eta}, \boldsymbol{\Theta}|\mathbf{x})$, i.e., $p(\boldsymbol{\eta}, \boldsymbol{\Theta}|\mathbf{x}) \approx q(\boldsymbol{\eta}, \boldsymbol{\Theta})$, which can be determined by solving the constrained optimization as follows:

$$q^*(\boldsymbol{\Theta}, \boldsymbol{\eta}) = \arg \max_{q(\boldsymbol{\Theta}, \boldsymbol{\eta}) \in \mathcal{Q}} F(q(\boldsymbol{\Theta}, \boldsymbol{\eta})), \text{ s.t. } \int q(\boldsymbol{\Theta}, \boldsymbol{\eta}) d\boldsymbol{\Theta} d\boldsymbol{\eta} = 1 \quad (10)$$

where $F(q(\boldsymbol{\eta}, \boldsymbol{\Theta})) = E_{\boldsymbol{\Theta}, \boldsymbol{\eta}}(\frac{p(\boldsymbol{\eta}, \boldsymbol{\Theta}|\mathbf{x})}{q(\boldsymbol{\eta}, \boldsymbol{\Theta})})$ is called the evidence lower bound up to the negative KL divergence, and $\mathcal{Q}$ represents the set of the potential variational family. In general, $\mathcal{Q}$ determines the complexity of VPD and the optimization problem (10) and further affects the computational burden and efficiency of the estimation of the model. Therefore, it is important to select

an appropriate $\mathcal{Q}$ that is simple enough to make (10) feasible and flexible enough to provide a good approximation of the true posterior. One such is the mean-field variational family (MFVF), which assumes the model parameters are mutually independent or blocking independent. With MFVF, it is further assumed that

$$q(\boldsymbol{\Theta}, \boldsymbol{\eta}) = \prod_{n=1}^N q(\eta_n) q(\alpha) q(\mu, \sigma^2) \quad (11)$$

where $q(\eta_n)$, $q(\alpha)$ and $q(\mu, \sigma^2)$ are the VPDs of η_n , α , and (μ, σ^2) . The VPD $q(\boldsymbol{\Theta}, \boldsymbol{\eta})$ contains three parts and has the same structure of $p(\boldsymbol{\eta}, \boldsymbol{\Theta}|\mathbf{x})$, which will improve the precision of approximation. Plugging the MFVF (11) into (10) and then solving (10) with the method of Lagrange multipliers [46], yields

$$F_1(q(\boldsymbol{\Theta}, \boldsymbol{\eta})) = F(q(\boldsymbol{\Theta}, \boldsymbol{\eta})) + \boldsymbol{\lambda}^T \left(\int (q(\boldsymbol{\Theta}, \boldsymbol{\eta}) - 1) d\boldsymbol{\Theta} d\boldsymbol{\eta} \right) \quad (12)$$

where the vector $\boldsymbol{\lambda} = (\{\lambda_{\eta_n}\}_{n=1}^N, \lambda_\alpha, \lambda_\mu, \lambda_{\sigma^2})$ is the Lagrange multiplier, which represents the marginal effect of $q(\boldsymbol{\Theta}, \boldsymbol{\eta})$ on the objective function.

The VB algorithm can be seen as a tradeoff between traditional EM and Bayesian methods regarding computational time and computational efficiency. The VB method approximates a complex Bayesian posterior by means of an easily handled proxy distribution. It transforms a Bayesian inference problem into a numerical optimization problem using the KL divergence as a link, i.e., it transforms a statistical inference problem based on stochastic simulation into a deterministic optimization problem. According to the discussion of Blei et al. [4], the optimization process of VB can be further divided into two iterative update steps for the VB posterior of the latent variables and the VB posterior of the parameters of interest. In analogy to the iterative process of the expectation maximization (EM) algorithm, Blei et al. [4] call the iterative process for the latent variable and the parameter of interest posterior the VB expectation (VB-E) and VB maximization (VB-M) steps, respectively.

1) *VB-E: Optimal Latent Variable*: Taking the functional derivative of (12) in terms of the VPD, $q(\boldsymbol{\eta})$, the logarithm of VPD of $\boldsymbol{\eta}$ can be obtained as

$$\begin{aligned} \ln(q(\boldsymbol{\eta})) &= \sum_{n=1}^N E_{\mu, \sigma^2} [\ln(p_\eta(\eta_n|\mu, \sigma^2))] \\ &+ \sum_{n=1}^N \sum_{m=1}^{M_n} E_\alpha [p_{x|\eta}(x_{n,m}|\alpha h_c, \eta_n)] + C. \end{aligned}$$

Due to the independent assumption, the logarithm of VPD of η_n is

$$\begin{aligned} \ln(q(\eta_n)) &= E_{\mu, \sigma^2} [\ln(p_\eta(\eta_n|\mu, \sigma^2))] \\ &+ \sum_{m=1}^{M_n} E_{q(\alpha)} [p_{x|\eta}(x_{n,m}|\alpha h_c, \eta_n)] + C. \end{aligned}$$

Using (7) and (8), we have

$$\ln(q(\eta_n)) = (w_1 w_2 - w_0 M h_c - 1) \ln(\eta_n) - \frac{y_{n,M} w_0}{\eta_n}$$

$$-w_2(\ln(\eta_n))^2/2 + C, \quad (13)$$

where $y_{n,M} = \sum_{m=1}^M x_{n,m}$, $w_0 = E_\alpha[\alpha]$, $w_1 = E_{\mu,\sigma^2}[\mu]$, and $w_2 = E_{\mu,\sigma^2}[\sigma^2]$, and the quantities $w_i, i = 0, 1, 2$ will be updated in the VB-M step. Then, the VPD of η_n can be obtained as $q(\eta_n) = g_0(\eta_n)/A_{\eta_n}$, provided by $g_0(\eta_n) = \eta_n^{w_1 w_2 - w_0 M h_c - 1} e^{-\frac{y_{n,M} w_0}{\eta_n} - \frac{w_2 (\ln(\eta_n))^2}{2}}$, $A_{\eta_n} = \int_0^{+\infty} g_0(\eta_n) d\eta_n$, where A_{η_n} is called the canonical function, which guarantees that $q(\eta_n)$ is a pdf.

Let $s = \ln(\eta_n) \sqrt{w_2/2}$. Plugging s into $g_0(\eta_n)$ yields

$$g_0(\eta_n) = q(s, y_{n,M} | w_0, w_1, w_2) e^{-s^2} \quad (14)$$

where

$$\begin{aligned} q(s, y_{n,M} | w_0, w_1, w_2) \\ = \sqrt{2/w_2} e^{s \sqrt{2/w_2} (w_1 w_2 - w_0 M h_c) - e^{-y_{n,M} w_0 s \sqrt{2/w_2}}} \end{aligned}$$

Notice that $g_0(\eta_n)$ has the same structure as the integrand in (5). Thus, A_{η_n} can also be evaluated by utilizing the GH rule introduced in Section II. Then, based on (13) and (14), the expectation for any continuous function $f(\eta_n)$ can be evaluated as

$$\begin{aligned} E_{\eta_n}(f(\eta_n)) &= \int_0^{+\infty} f(\eta_n) q(\eta_n) d\eta_n \\ &= \frac{1}{A_{\eta_n}} \int_{-\infty}^{+\infty} f(\exp(s \sqrt{2/w_2})) e^{-s^2} \\ &\quad \times q(s, y_{n,M} | w_0, w_1, w_2) ds \end{aligned} \quad (15)$$

which can also be computed by GH quadrature numerically.

2) *VB-M: Optimal Model Parameters:* Taking the functional derivative of (12) in terms of the VPD $q(\Theta)$, the logarithm of VPD of Θ can be obtained as

$$\begin{aligned} \ln(q(\alpha, \mu, \sigma^2)) &= \sum_{n=1}^N [E_{\eta_n} [\ln(p(\eta_n | \mu, \sigma^2))]] \\ &\quad + \sum_{n=1}^N \sum_{m=1}^M [E_{q(\eta_n)} [\ln(p(x_{n,m} | \eta_n, \alpha))]] \\ &\quad + \ln(p(\alpha)) + \ln(p(\mu, \sigma^2)) + C. \end{aligned}$$

Notice that the terms of α and the terms of (μ, σ^2) are well separated. Thus, given the VPDs $q(\eta_n), n = 1, 2, \dots, N$, the expressions of the VPDs, $q(\alpha)$ and $q(\mu, \sigma^2)$ can be derived sequentially.

1) *VPD of α :* The logarithm of VPD of α is given by

$$\begin{aligned} \ln(q(\alpha)) &= \sum_{n=1}^N \sum_{m=1}^M [E_{q(\eta_n)} [\ln(p(x_{n,m} | \eta_n, \alpha))]] \\ &\quad + \ln(p(\alpha)) + C. \end{aligned}$$

Then, plugging (7) and (8) into (22), we have

$$\begin{aligned} \ln(q(\alpha)) &= -(h_c(Mw_3 - w_5) + w_4)\alpha + \alpha h_c N M \ln(\alpha) \\ &\quad - N M \ln(\Gamma(\alpha h_c)) + \ln(p(\alpha)) + C \end{aligned} \quad (16)$$

where $w_3 = \sum_{n=1}^N E_{\eta_n}[\ln(\eta_n)]$, $w_4 = \sum_{n=1}^N y_{n,M} E_{\eta_n}[\eta_n^{-1}]$, $w_5 = \sum_{n=1}^N \sum_{m=1}^M \log(x_{n,m})$, and $E_{\eta_n}[\ln(\eta_n)]$ and $E_{\eta_n}[\eta_n^{-1}]$ can be updated with (15). Taking the exponential transformation on both sides of (16) leads to

$$q(\alpha) \propto p(\alpha) \frac{\alpha^{a\alpha}}{\Gamma(\alpha h_c)^b} \exp(-c\alpha) \quad (17)$$

where $a = h_c N M$, $b = N M$, and $c = h_c(Mw_3 - w_5) + w_4$. For the convenience of the mathematical derivation, it is common to use a conjugate prior if it is available. From the definition of conjugate prior [47], the VPD $q(\alpha)$ and $p(\alpha)$ should be in the same distribution family. In view of this, we assign $p(\alpha)$ as

$$p(\alpha) \propto \frac{\alpha^{a_0}}{\Gamma(\alpha h_c)^{b_0}} \exp(-c_0 \alpha) \quad (18)$$

where a_0, b_0 , and c_0 are hyperparameters in prior $p(\alpha)$. Then, plugging (18) into (17) and calculating the normalization constant, we have

$$q(\alpha) = g_\alpha(\alpha)/A_\alpha \quad (19)$$

provided by $g_\alpha(\alpha) = \frac{\alpha^{a_1 \alpha}}{\Gamma(\alpha h_c)^{b_1}} \exp(-c_1 \alpha)$, $A_\alpha = \int_{\Omega_\alpha} g_\alpha(\alpha) d\alpha$, where $a_1 = a_0 + a$, $b_1 = b_0 + b$, and $c_1 = c_0 + c$. However, it is difficult to evaluate the VPD mean and variance of α through (19), as the integration of $g_\alpha(\alpha)$ may be not numerically stable. Alternatively, the Laplace approximation is applied on (19). The VPD mean and variance of α can then be approximated as

$$\begin{aligned} m(\alpha) &\approx \arg \min_{\alpha} \{a_1 \log(\alpha) - b_1 h_c \psi(\alpha h_c) + a_1 - c_1\} \\ v(\alpha) &\approx \frac{1}{b h_c^2 \psi'(\alpha h_c) - \frac{a_1}{m(\alpha)}}. \end{aligned}$$

2) *Joint VPD of blocking parameters (μ, σ^2) :* Based on (16), the logarithm joint VPD of (μ, σ^2) can be written as

$$\begin{aligned} \ln(q(\mu, \sigma^2)) &= \sum_{n=1}^N [E_{q(\eta_n)} [\ln(p(\eta_n | \mu, \sigma^2))]] \\ &\quad + \ln(p(\mu, \sigma^2)) + C. \end{aligned} \quad (20)$$

Then, substituting (7) and (8) into (20), and taking exponential of the result, we have

$$q(\mu, \sigma^2) \propto p(\mu, \sigma^2) (\sigma^2)^{-N/2} e^{-\frac{w_6 - w_3^2/N}{2\sigma^2} - \frac{N(\mu - w_3/N)^2}{2\sigma^2}} \quad (21)$$

where $w_6 = \sum_{n=1}^N E_{q(\eta_n)}[(\ln(\eta_n))^2]$, and $E_{\eta_n}[(\ln(\eta_n))^2]$ can be evaluated through (15). (21) happens to be the kernel of the normal inverse gamma (NIGa) density function [48], only if $p(\mu, \sigma^2)$ is also a NIGa distribution. In general, the random variables (X, Y) are said to be NIGa distribution $\text{NIGa}((x, y); \rho, \varsigma, \nu, \xi)$ with pdf

$$p(x, y) = \frac{\sqrt{\varsigma}}{\sqrt{y} \sqrt{2\pi}} \frac{\xi^\nu}{\Gamma(\nu)} y^{-\nu-1} e^{-\frac{2\xi + \varsigma(x-\rho)^2}{2y}} \quad (22)$$

provided by $x \in (-\infty, +\infty)$ and $y \in (0, +\infty)$, where $\nu > 0, \xi > 0, \varsigma > 0$, and $\rho \in (-\infty, +\infty)$ are the parameters. From the definition of NIGa distribution [49], (22) can be decomposed into the product of a conditional normal distribution of μ with mean ρ and variance σ^2/ς given σ^2 and an inverse gamma distribution of σ^2 with shape parameter ν and scale parameter ξ . Thus, we assume that the joint prior for blocking parameters (μ, σ^2) is

$$p(\mu, \sigma^2) = NIGa((\mu, \sigma^2); \rho_0, \varsigma_0, \nu_0, \xi_0). \quad (23)$$

Plugging (23) into (21), the joint VPD of (μ, σ^2) can be obtained

$$q(\mu, \sigma^2) = NIGa((\mu, \sigma^2); \rho_1, \varsigma_1, \nu_1, \xi_1) \quad (24)$$

where $\rho_1 = \frac{1}{\varsigma_1}[\varsigma_0\rho_0 + w_3]$, $\varsigma_1 = \varsigma_0 + N$, $\nu_1 = \nu_0 + \frac{N}{2}$, $\xi_1 = \xi_0 + \frac{1}{2}[\varsigma_0\rho_0^2 + w_6 - \varsigma_1\rho_1^2]$. Then, according to the properties of NIGa distribution [48], the VPD mean and variance of μ and σ^2 are given by

$$m(\mu) = \rho_1, \quad v(\mu) = \frac{\xi_1}{\varsigma_1(\nu_1 - 1)}$$

$$m(\sigma^2) = \frac{\xi_1}{\nu_1 - 1}, \quad v(\sigma^2) = \frac{\xi_1^2}{(\nu_1 - 1)^2(\nu_1 - 2)}$$

and the first inverse moment of σ^2 is equal to ν_1/ξ_1 , which are required in VB-E step.

With the updating equations of VB-E and VB-M for the current model, iterations can be conducted between them until there is no significant improvement in terms of the estimated variance, i.e.

$$\varepsilon(\Theta^{(s)}) = \|v(\Theta^{(s)}) - v(\Theta^{(s-1)})\|_1 \leq \varepsilon_0 \quad (25)$$

where $\|\cdot\|_1$ is the absolute-value norm, ε_0 is a prespecified value that determines the desired accuracy.

Remark 1: The key to the successful and efficient working of the VB algorithm is the assumption of an MFVF and the conjugate character of the model parameters. With the MFVF, we can obtain an approximate Bayesian posterior based on the K-L divergence by means of a Lagrangian optimization algorithm. The conjugate character of the model parameters allows us to obtain an approximate Bayesian posterior with an analytic expression. For nonconjugate model parameters, we can obtain the corresponding variational posterior by Laplace approximation based on the natural conjugate prior. A model with more conjugate parameters will also be more computationally efficient and accurate. The computational efficiency is higher because it contains fewer numerical optimization processes, while the accuracy is higher because the conjugate part has a similar structure in the posterior of VB and Bayesian algorithms. If η is set to obey other skewed random distributions defined on the real axis, the model will contain more nonconjugate parameters, thus reducing the overall computational efficiency and accuracy.

B. Determination of the Initial Values

In the implementation of the VB algorithm, the choice of initial values of the parameters is especially important because it can affect the convergence time of the Laplace approximation

of parameter α and its ability to find an excellent solution. Thus, we should choose initial points carefully for the parameters Θ in the proposed model. In this section, the initial values of the parameters Θ would be determined through the following procedure.

- 1) For unit n , the mean and variance of increments X_n can be calculated as $\bar{x}_n = \frac{1}{M} \sum_{m=1}^M x_{n,m}$ and $V_n = \frac{1}{M-1} \sum_{m=1}^M (x_{n,m} - \bar{x}_n)^2$, respectively.
- 2) Since increments X_n follows $Ga(\alpha h_c, \eta_n/\alpha)$ with mean $\eta_n h_c$ and variance $\eta_n^2 h_c/\alpha$, η_n and α can be roughly estimated as $\bar{x}_n/h_c$ and $\bar{x}_n^2/(V_n h_c)$.
- 3) With $\{\eta_n\}_{n=1}^N$, the initial values of parameters μ and σ^2 can be roughly determined as $\bar{\eta}_n = \frac{1}{N} \sum_{n=1}^N \ln(\eta_n)$ and $V_{\ln(\eta_n)} = \frac{1}{N-1} \sum_{n=1}^N (\ln(\eta_n) - \bar{\eta}_n)^2$.

C. Confidence Interval

With the above iterative VB estimator, the VPD mean $m(\theta_i)$ and variance $v(\theta_i)$ for each $\theta_i \in \Theta$ can be obtained. Then, a $100(1 - \gamma)\%$ normal-approximation confidence intervals (CI) of θ_i can be taken as

$$m(\theta_i) \pm Z_{1-\gamma/2} \sqrt{v(\theta_i)} \quad (26)$$

where Z_γ is the γ th quantile of the standard normal distribution. However, (26) is inadequate because the VB approach usually underestimates the spread of the posterior and further results in a smaller value of estimate of the variance than that with the maximum likelihood (ML) approach [4]. To deal with this problem, a refined estimation of CI for VB estimator was discussed and proposed by Wang and Titterton [50], in which the variance of VPD is computed by the Fisher information matrix under the original likelihood function. When the observations are available, the Fisher information matrix can be further replaced with the observed Fisher information matrix [45]. With this idea, the observed Fisher information matrix for the current model is given by

$$\hat{I}_\Theta = -\frac{\partial^2 \log(\ell(\Theta))}{\partial \Theta^T \partial \Theta} \quad (27)$$

where

$$\begin{aligned} \ell(\Theta) = & -NM \log(\Gamma(\alpha h_c)) - NM \alpha h_c \log(\alpha) \\ & + \sum_{n=1}^N \sum_{m=1}^M (\alpha h_c - 1) \log(x_{n,m}) - \frac{N}{2} \log(\pi) \\ & + \sum_{n=1}^N \log \left[\int_{-\infty}^{\infty} q(z, y_{n,M}, \Theta) \exp(-z^2) dz \right] \end{aligned}$$

provided by $z = \frac{\log(\eta_n - \mu)}{\sqrt{2\sigma^2}}$ and

$$q(z, y_{n,M}, \Theta) = e^{-\alpha h_c M (\sqrt{2\sigma^2} z + \mu) - \alpha y_{n,M} e^{-\sqrt{2\sigma^2} z - \mu}}. \quad (28)$$

The detail derivatives for (27) are listed in the Appendix.

D. Benchmark Estimators

Two benchmark estimators, the Bayesian estimator and the EM estimator, will be considered in this section, which are,

respectively, compared with the proposed VB estimator in numerical simulations and case studies.

The EM algorithm is commonly used to conduct ML estimation (MLE) for the intractable likelihood functions, especially for models with missing data [51]. For the proposed gamma degradation model, let the hidden variables η be the missing data. The EM algorithm can then be used to obtain the MLEs of Θ , which works as follows.

E-step: Given the estimators $\Theta^{(s)}$ after the s th iteration, the required moments of η_n for subsequent M-step are calculated as

$$\begin{aligned} E_{1n}^{(s)} &= E_{q_1(\eta_n)}[\ln(\eta_n)]|_{\Theta=\Theta^{(s)}} \\ E_{2n}^{(s)} &= E_{q_1(\eta_n)}[1/(\eta_n)]|_{\Theta=\Theta^{(s)}} \\ E_{3n}^{(s)} &= E_{q_1(\eta_n)}[(\ln(\eta_n))^2]|_{\Theta=\Theta^{(s)}} \end{aligned} \quad (29)$$

where

$$q_1(\eta_n) \propto \eta_n^{\mu^{(s)}/(\sigma^2)^{(s)} - \alpha^{(s)} M h_c - 1} e^{-\frac{\alpha^{(s)} y_{n,M}}{\eta_n} - \frac{(\ln(\eta_n))^2}{2(\sigma^2)^{(s)}}} \quad (30)$$

with $\alpha^{(s)}$, $\mu^{(s)}$, and $(\sigma^2)^{(s)}$ from the M-step below. It is noted that (30) can also be converted to GH quadrature, which is then utilized to calculate $E_{dn}^{(s)}$, $d = 1, 2, 3$ in the E-step.

M-step: Updating the parameters Θ with $E_{dn}^{(s)}$, $d = 1, 2, 3$ as

$$\begin{aligned} \alpha^{(s+1)} &= \arg \min_{\alpha \in \Omega_\alpha} \left\{ |NM[h_c(\ln(\alpha) + 1) - \psi(\alpha h_c)] \right. \\ &\quad \left. + h_c \left(w_5 - M \sum_{n=1}^N E_{1n}^{(s)} \right) - w_6 \sum_{n=1}^N E_{2n}^{(s)} \right\} \\ \mu^{(s+1)} &= \frac{1}{N} \sum_{n=1}^N E_{1n}^{(s)} \\ (\sigma^2)^{(s+1)} &= \frac{1}{N} \sum_{n=1}^N \left[E_{3n}^{(s)} - (\mu^{(s+1)})^2 \right]. \end{aligned} \quad (31)$$

The implementation of the EM algorithm starts with an initial value of $\Theta^{(0)}$ and then iterates (29) and (31) until some convergence criterion is satisfied.

The Bayesian approach can capture the features of posteriors of the parameters of interest by gathering the information from data and priors. However, it is a challenging task to get the closed form for each marginal posterior in the Bayesian analysis. Alternatively, MCMC methods, in particular, the Gibbs sampler can be used to obtain the posterior samples from a set of Markov chains with respect to the parameters of interest and nuisance parameters (if existing), and the iteration is terminated until all chains are stable and mixed well. For the current RGA degradation model, the MCMC-based framework is as follows.

- 1) With current Bayesian estimator $\Theta^{(s)}$, generate $\eta_n^{(s+1)}$ from conditional posterior of η_n

$$p_2(\eta_n|\Theta^{(s)}) = \prod_{m=1}^M p_{x|\eta}(x_{n,m}|\alpha^{(s)}). \quad (32)$$

- 2) With current Bayesian estimator $\eta^{(s+1)}$, sample $\Theta^{(s+1)}$ from conditional posteriors of α

$$p_2(\alpha|\eta^{(s+1)}) = \prod_{n=1}^N \prod_{m=1}^M p_{x|\eta}(x_{n,m}|\eta_n^{(s+1)}, \alpha)p(\alpha) \quad (33)$$

and the conditional posterior of (μ, σ^2)

$$p_2(\mu, \sigma^2|\eta^{(s+1)}) = \prod_{n=1}^N p_\eta(\eta_n^{(s+1)}|\mu, \sigma^2)p(\mu, \sigma^2) \quad (34)$$

where $p(\alpha)$ and $p(\mu, \sigma^2)$ are the prior of α and (μ, σ^2) , which are the same as the setup in the VB algorithm. The implementation of the Bayesian method starts with the setting of the initial values $\Theta^{(0)}$ and $\eta^{(0)}$ and generates the posterior sample based on (32)–(34) iteratively until all chains are stable and mixed well.

IV. NUMERICAL SIMULATIONS

In the simulation experiments, two objectives will be assessed. The first is to verify the computational accuracy of the developed VB algorithm by comparing it with EM and Bayesian algorithms. The other is the computing efficiency of three algorithms. Some running configurations for the VB, EM, and MCMC algorithms are set as follows.

- 1) VB algorithm: The hyperparameters in priors $p(\alpha)$ and $p(\mu, \sigma^2)$ can be determined according to the historical data or experts' experiences, while weakly informative priors are usually selected in absence of these information. For a fair comparison with the EM algorithm, the hyperparameters are chosen as $a_0 = 1, b_0 = 1, c_0 = 1, \xi_0 = 1, \varsigma_0 = 1, \rho_0 = 1$, and $\nu_0 = 0$, which implies weak information for priors.
- 2) EM algorithm: The absolute-value norm of the estimators of the last two iterations will be utilized as the stop criterion.
- 3) MCMC algorithm: Identical setups of the hyperparameters will be utilized for the purpose of comparison. The number of Markov chains in the posterior sampling is three, and the number of total iterations per chain is 20 000, with the first half of iterations as the burn-in period.
- 4) Common settings: The desired accuracy for VB and EM is $1e-6$. All algorithms are implemented through R software, and the running environment is a Windows10 desktop computer with a 3.5 GHz Intel(R) Xeon(R) CPU E3-1241 v3 processor and 16 GB of random access memory.

The sample size N is chosen as 15, 30, and 60, and for each N , the number of measurements $M = 10$ and 20. For each combination (N, M) , generate 2000 degradation samples $\mathbf{y}$ randomly from the proposed gamma degradation model with parameters $\Theta = (\alpha, \mu, \sigma^2) = (10, -1, 0.01)$. For each sample $\mathbf{y}$, the estimates of Θ are obtained via the VB, EM, and MCMC algorithms. Thus, we obtain 2000 estimates for each algorithm. Then, based on these estimates, the average and mean-squared errors (MSEs) are computed. The results are listed in Table I. We also calculate the average CPU running time in terms of seconds ($\bar{t}$) for each algorithm, which is shown in Fig. 3. In addition,

TABLE I
AVERAGES AND MSEs OF THE ESTIMATES UNDER DIFFERENT ALGORITHMS
AND SAMPLE SIZES

(N, M)	Method	$\alpha = 10$		$\mu = -1$		$\sigma^2 = 0.01$	
		MSE	Mean	MSE ($\times 10^{-3}$)	Mean	MSE ($\times 10^{-5}$)	Mean
(15, 10)	VB	3.5689	10.2939	2.1490	-0.9921	12.7358	0.0118
	EM	3.7443	10.4057	2.4983	-0.9990	13.8214	0.0118
	Bayes	1.2705	10.2116	1.2181	-1.0007	9.3512	0.0179
(15, 20)	VB	3.0234	10.3048	1.3635	-0.9973	8.9276	0.0118
	EM	3.4744	10.3870	1.3836	-1.0011	9.3441	0.0114
	Bayes	0.7495	10.1547	1.1617	-1.0011	5.1783	0.0116
(30, 10)	VB	2.4774	10.2291	1.1519	-0.9990	4.1917	0.0097
	EM	3.4369	10.3056	1.1763	-0.9986	3.0103	0.0108
	Bayes	0.6956	10.2142	0.8215	-1.0083	1.8619	0.0103
(30, 20)	VB	2.2640	10.1209	0.8165	-0.9974	2.4172	0.0097
	EM	2.6755	10.1871	0.8234	-0.9988	2.6170	0.0104
	Bayes	0.3813	10.0223	0.5394	-1.0001	1.1418	0.0102
(60, 10)	VB	2.0031	10.0133	0.4217	-1.0121	1.1013	0.0104
	EM	2.042	10.0147	0.4325	-1.0137	1.2131	0.0107
	Bayes	0.2032	10.0125	0.3142	-1.0015	0.9324	0.0102
(60, 20)	VB	2.0017	10.093	0.3627	-1.0097	1.0121	0.0098
	EM	2.0014	10.0104	0.3674	-1.0117	1.1374	0.0103
	Bayes	0.1747	10.0041	0.2873	-1.0009	0.7213	0.0101

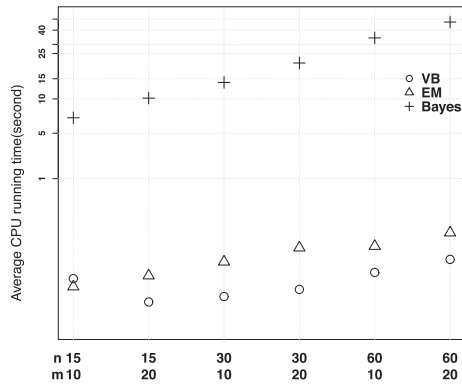


Fig. 3. Average CUP running time for proposed model under different algorithms and sample sizes.

95% coverage probabilities (CPs) and the related interval lengths (ILs) are also calculated, which can be found in Table II. From the two tables and the figure of the average CPU running time of considered three algorithms, some results are found as follows.

- 1) For all cases, the averages of the estimates are close to the true values, and the differences between the averages and the true values get smaller and smaller as the sample size increases.
- 2) As the sample size increases, MSEs show a trend of gradual decline overall for each simulation setting. The MSEs under VB and EM are smaller than the MSEs under Bayesian, and the differences of MSEs under Bayesian and other two methods get smaller and smaller as the sample size increases.
- 3) As expected, the ILs decrease with the increase in the sample size for all algorithms. Moreover, the difference in ILs between VB and Bayesian estimators also displays a decreasing tendency with the increase in the sample size.
- 4) For CP, the performance of the VB algorithm is slightly better than that of the EM algorithm because the CPs are closer to the nominal level of 0.95. While the Bayesian

TABLE II
95% CPs AND ILs OF INTERVAL ESTIMATES UNDER DIFFERENT ALGORITHMS
AND SAMPLE SIZES

(N, M)	Method	$\alpha = 10$		$\mu = -1$		$\sigma^2 = 0.01$	
		CPs	ILs	CPs	ILs	CPs	ILs
(15, 10)	VB	0.9123	3.9051	0.9234	0.7675	0.9014	0.7234
	EM	0.9136	4.5865	0.9216	0.7921	0.8945	0.8653
	Bayes	0.9208	3.8839	0.9307	0.1688	0.9134	0.5053
(15, 20)	VB	0.9293	2.7575	0.9356	0.6060	0.9345	0.4209
	EM	0.9294	3.2482	0.9341	0.7459	0.9301	0.5678
	Bayes	0.9406	2.7325	0.9406	0.1471	0.9406	0.3847
(30, 10)	VB	0.9323	3.8864	0.9245	0.5655	0.9471	0.4515
	EM	0.9289	3.9691	0.9223	0.7319	0.9412	0.5317
	Bayes	0.9501	3.3626	0.9307	0.1092	0.9505	0.2628
(30, 20)	VB	0.9457	2.7569	0.9451	0.4971	0.9487	0.3124
	EM	0.9470	2.7751	0.9412	0.5777	0.9465	0.4018
	Bayes	0.9503	2.2675	0.9504	0.0947	0.9507	0.1987
(60, 10)	VB	0.9473	2.6321	0.9462	0.4737	0.9503	0.2132
	EM	0.9469	2.7747	0.9451	0.5153	0.9495	0.3314
	Bayes	0.9505	1.8278	0.9507	0.0879	0.9510	0.1637
(60, 20)	VB	0.9489	2.4143	0.9475	0.4687	0.9504	0.2047
	EM	0.9475	2.6032	0.9457	0.5096	0.9502	0.3176
	Bayes	0.9509	1.8021	0.9511	0.0813	0.9513	0.1521

algorithm performs better than the other two algorithms in the cases of small sample sizes, e.g., (15, 10) and (15, 20).

- 5) Compared with EM and Bayesian, VB shows significant computing advantages, the average running time of which is only one-hundredth or even one-thousandth of the Bayesian under some situations.

In general, the estimation accuracy of the three algorithms gradually increases with the increase in the sample size. The Bayesian algorithm is closer to the true posterior estimate and has the highest estimation accuracy. However, we also need to see that the Bayesian algorithm becomes less and less efficient when the sample size increases. The traditional EM algorithm improves the computational efficiency but sacrifices some of the computational accuracy. In contrast, the VB algorithm can be seen as a tradeoff between the Bayesian algorithm and the EM algorithm, which not only outperforms the other two algorithms in terms of computational efficiency but also improves the computational accuracy of the EM algorithm.

V. CASE STUDY

A. Laser Degradation Data

The Gallium arsenide (GaAs) laser data [45] is utilized to illustrate the proposed RGa degradation model and verify the developed VB algorithm. The operating current is utilized as the degradation characteristic for the laser device, which shows an increasing tendency over the life of the laser device for maintaining the constant light output. When the operating current of the laser device exceeds a critical level, the laser device is considered to have suffered a pseudofailure. Fig. 4 displays the degradation path of 15 GaAs laser specimens, measured with frequency 250 h up to 4000 h under 80 °C.

Two objectives are first verified before fitting the laser data with the RGa degradation model. The first one is to check whether the unit-to-unit variability of the degradation rate can be captured by log-normal distribution. The other is to study the

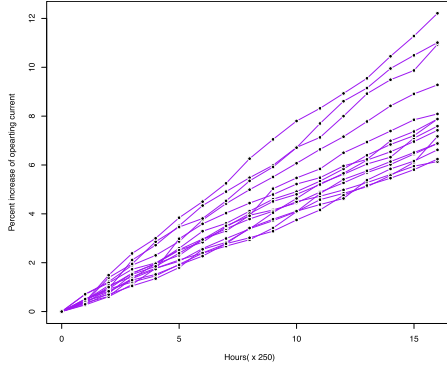


Fig. 4. Degradation paths of laser data.

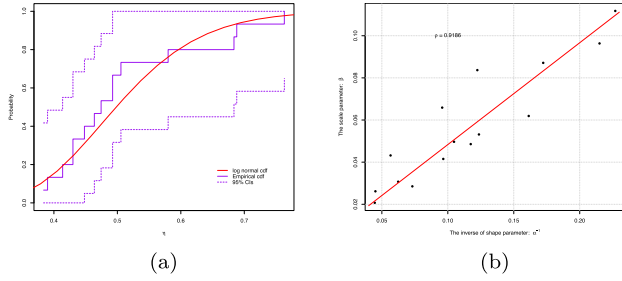


Fig. 5. (a) Estimated cdf of the log-normal distribution and empirical cdf with 95% CIs for $\hat{\eta}$ and (b) the scatter plot of the inverse of estimated shape and the scale parameter in RGa process, and their regression line and correlation coefficient ρ based on laser degradation data.

dependence of shape and scale parameters in the RGa process for the laser data. For the first objective, the empirical estimator $\hat{\eta} = \{\hat{\eta}_n = \frac{y_{nM}}{t_{nM}}\}_{n=1}^{15}$ is first calculated, which is then utilized to calculate the parameters in log-normal distribution with the formula proposed in (2). With simple calculation, we obtain $\hat{\eta} \sim LN(-0.697, 0.045)$. Fig. 5(a) shows the empirical cdf with 95% CIs and the estimated cdf of log-normal distribution, which indicates the degradation rate $\hat{\eta}$ can be well captured by the log-normal distribution. Besides, the Anderson–Darling test is adopted for checking the goodness-of-fit of the log-normal distribution. The p -value is 0.633, which shows a satisfactory fitting with the log-normal distribution. Next, we explore the dependence of shape and scale parameters in the RGa process for the laser data. The empirical estimators of shape and scale parameters are $\hat{\alpha}, \hat{\beta} = \{\hat{\alpha}_n = \frac{\hat{\eta}_n^2}{v(\mathbf{x}_n/\sqrt{h_c})}, \hat{\beta}_n = \frac{\hat{\eta}_n}{\hat{\alpha}_n}\}_{n=1}^{15}$. Fig. 5(b) shows the regression curve of $\hat{\alpha}^{-1}$ and $\hat{\beta}$, which suggests an extreme positive dependence between the estimated scale and the inverse of estimated shape parameters.

According to the above analysis, we then fit the laser data with the proposed RGa degradation model under the developed VB algorithm. To illustrate the computational efficiency and fitting accuracy, the proposed RGa degradation model is also evaluated under the EM and Bayesian approaches. Table III shows the estimates and the corresponding 95% CIs of the parameters and the running time under each algorithm. From Table III, we can obtain that there are no significant differences in the estimates of parameters under different approaches. However, the VB

TABLE III
ESTIMATES (ESTS) AND 95% CIs OF THE PARAMETERS AND THE COMPUTATION TIME UNDER DIFFERENT METHODS FOR THE LASER DATA

Method	α		μ		σ^2		time
	Est	95% CIs	Ests	95% CIs	Ests	95% CIs	
VB	9.682	(8.161, 11.792)	-0.696	(-0.854, -0.549)	0.059	(0.019, 0.141)	0.1842
EM	9.713	(8.159, 11.802)	-0.696	(-0.856, -0.550)	0.058	(0.019, 0.143)	0.7451
Bayes	9.699	(8.478, 11.702)	-0.694	(-0.820, -0.561)	0.057	(0.021, 0.138)	12.381

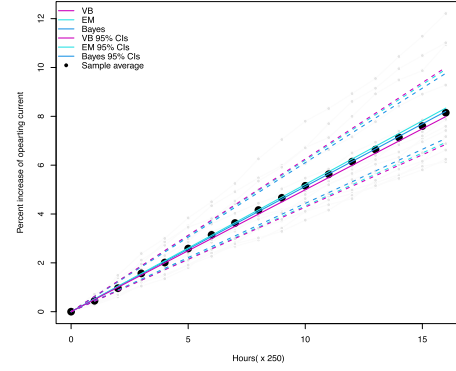


Fig. 6. Sample average and the estimated mean paths based on different methods for the laser data.

algorithm has an obvious advantage in terms of CPU running time, which is less than the EM algorithm and much less than the Bayesian algorithm. Fig. 6 shows the sample average value, the estimated mean degradation path ($\hat{\eta}t$), and the corresponding 95% CIs for each algorithm. From Fig. 6, we can see that the sample average points can be well fitted by the estimated mean degradation path for all algorithms. Model selection is performed under five candidate models: 1) gamma process, 2) inverse Gaussian process, 3) Wiener process, 4) the homogeneous gamma process with random effects (Tsai et al. [26]), and 5) the proposed RGa process. For notation convenience, we denote the proposed RGa model and the model of Tsai et al. [26] as RGaLn and GaGa, respectively. Since there is no significant difference between the computed results of the given models under the EM and VB algorithms, we apply the estimates under the EM algorithm in the model selection. Therefore, we can use Akaike's information criteria (AIC) for the model selection. In addition, it is noted that the Bayesian information criteria (BIC) involve the number of valid samples while the VB algorithm is a deterministic approximation method. Therefore, BIC is not applicable to the model selection under the VB algorithm. The results are listed in Table IV. From Table IV, we can see that the AIC values based on the two random effect models: 1) GaGa and 2) RGaLn, are significantly smaller than the other three models. Therefore, it is reasonable to take the random effect into consideration for the laser degradation data. However, there are no significant differences between the AICs of RGaLn and GaGa models. Fig. 7 shows empirical cdf, and cdfs of lifetime under GaGa and RGaLn models, with threshold level $\omega = 6$ [17]. The empirical cdf in Fig. 7 is obtained the nonparametric estimation of the product's pseudofailure times using the median ranks estimator [45]. From Fig. 7, we see that both GaGa and RGaLn models can fit the empirical cdf well.

TABLE IV
VALUES OF THE LOG-LIKELIHOODS AND AICs BASED ON DIFFERENT MODELS
FOR THE LASER DEGRADATION DATA

Model	log-likelihood	AIC
RGaLn	93.291	-180.581
GaGa	93.738	-181.476
RGa	69.609	-135.219
Gamma	69.609	-135.219
Inverse Gaussian	73.031	-148.062
Wiener	45.563	-87.126

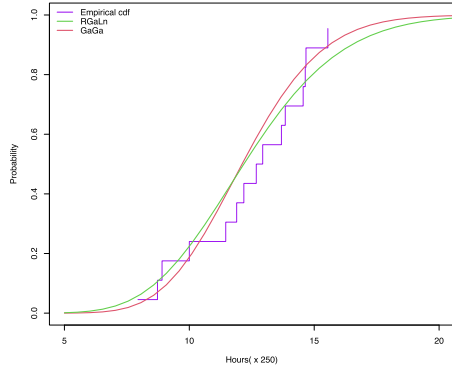


Fig. 7. Empirical cdf and the estimated cdf of lifetime based on RGaLn and GaGa models for the laser data.

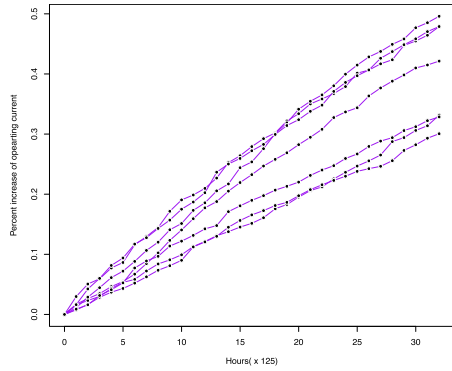


Fig. 8. Degradation paths of the Device-B data.

B. Device-B Data

The integrated circuit devices have been widely mounted on many highly reliable products, the power output of which will decrease over time. The device-B data [45] consist of a collection of degradation data of a particular integrated circuit device. Fig. 8 depicts the degradation paths of 7 device-B specimens, each of which is measured with a frequency of 125 h up to 4000 h under 150 °C.

Fig. 9(a) displays the empirical cdf with 95% CIs and the estimated cdf of log-normal distribution for the empirical estimation of the degradation rate $\hat{\eta}$, which suggests that the log-normal distribution performs well. The p -value of the Anderson–Darling test of the log-normal is 0.883. Thus, it is reasonable to model

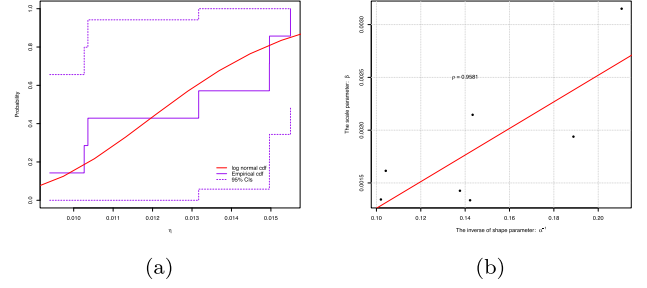


Fig. 9. (a) Estimated cdf of the log-normal distribution and empirical cdf with 95% CIs for $\hat{\eta}$ and (b) the scatter plot of the inverse of estimated shape and the scale parameter in RGa process, and their regression line and correlation coefficient ρ based on the device-B data.

TABLE V
ESTIMATES (ESTS) AND 95% CIs OF THE MODEL PARAMETERS AND THE
COMPUTATION TIME UNDER DIFFERENT METHODS FOR THE DEVICE-B DATA

Method	α		μ		σ^2		time
	Est	95% CI	Est	95% CI	Est	95% CI	
VB	7.681	(6.381, 9.171)	-4.384	(-4.653, -4.119)	0.112	(0.021, 0.195)	0.643
EM	7.679	(6.380, 9.162)	-4.384	(-4.653, -4.117)	0.110	(0.023, 0.97)	1.423
Bayes	7.689	(6.376, 9.207)	-4.385	(-4.655, -4.120)	0.116	(0.019, 0.201)	20.506

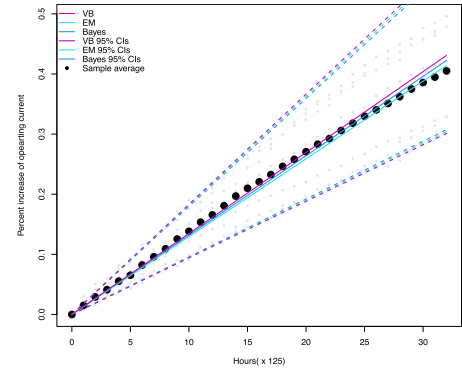


Fig. 10. Sample average and the estimated mean paths based on different for the device-B data.

the unit-to-unit variability with the log-normal distribution. Fig. 9(b) shows the regression line of the estimated shape and the inverse of scale parameters as well as the correlation coefficient ρ , which indicates a very strong positive dependence between them.

Then, the device-B data are fitted with the proposed RGa degradation model under VB, EM, and Bayesian methods, respectively, and the results are listed in Table V. From Table V, we conclude that the VB algorithm has a significant computing advantage and satisfactory precision. Fig. 10 displays the sample average value and the estimated mean degradation path ($\hat{\eta}t$) for each algorithm. From Fig. 10, we can conclude that the sample average points can be well fitted by the estimated mean degradation path for all algorithms.

Next, the device-B data are fitted by five different models, and the values log-likelihoods and AICs are listed in Table VI. From Table VI, it is concluded that the proposed RGaLn is the optimal model for the device-B data. Noted that the AIC of the RGaLn

TABLE VI
VALUES OF LOG-LIKELIHOODS AND AICs BASED ON DIFFERENT MODELS FOR
THE DEVICE-B DATA

Model	log-likelihood	AIC
RGaLn	893.009	-1780.018
GaGa	888.156	-1770.312
RGa	876.595	-1749.190
Gamma	876.595	-1749.190
Inverse Gaussian	851.529	-1699.058
Wiener	881.747	-1759.494

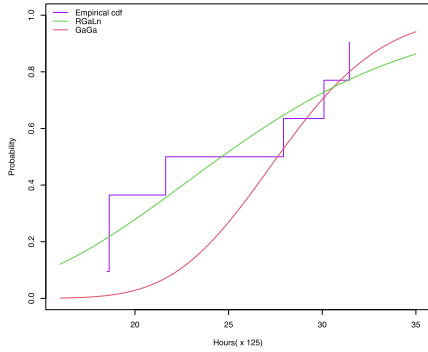


Fig. 11. Empirical cdf and the estimated cdf of lifetime based on RGaLn and GaGa models the Device-B data.

model is calculated with its parameters estimation under the EM algorithm. Finally, as a comparison, we draw the empirical cdf and the estimated cdfs of the lifetime with degradation threshold level $\omega = 0.3$ under GaGa and RGaLn models in Fig. 11, from which we can see that the goodness-of-fit based on the proposed RGaLn model is much better than that based on the GaGa model.

VI. CONCLUSION

In this article, an RGa degradation model is developed, which has a clear physical interpretation on parameters and maintains the dependence of degradation mean and variance as the traditional gamma degradation model, and random effects can be incorporated into the degradation rate directly. But more importantly, the shape parameter and blocking parameters in the random effects part are separated, which makes VB inference feasible. In the development of the VB algorithm, GH quadrature is utilized to evaluate the numerical integration, and Laplace approximation is used to yield the closed-form variational mean and variance for the intractable VPD of shape parameter. In consideration of the inadequacy of interval estimates of the VB approach, an adjusted interval estimate is developed by using the observed Fisher information matrix. Numerical experiments show the computational efficiency and estimation accuracy of the VB algorithm. The laser degradation data and device-B data are analyzed for illustration, and the results indicate that the proposed gamma degradation model could provide a satisfactory fitting.

Although the VB algorithm has been widely used in several fields, it still has many challenges and limitations in practical

applications. For the nonconjugate part of the model parameters, it is difficult for us to obtain the analytic solutions of the parameters under the mean-field VB inference. Moreover, when the model parameters have high dimensionality or high correlation of latent variables, it is difficult to obtain satisfactory estimation accuracy based on the mean-field VB method. For example, for a traditional degenerate model of a gamma process with random effects [26], the parameters of the random effects part do not have an analytic VB solution. Although Stan [52] integrates the VB algorithm for engineering applications, it directly approximates the model parameters of immediate interest with a multivariate normal distribution. When the model parameters are skewed, the parameter estimates are subject to large biases. To address these limitations in the application of the VB algorithm, we give an alternative VB inference strategy. In mean-field variational inference, we assume that the model parameters are block independent, preserving the correlation between parameters as much as possible. Then, we perform VB inference on the conjugate part of the model parameters, which not only obtains the analytical solution of the conjugate part of the model parameters but also reduces the dimensionality of the model parameters. Finally, we then perform approximate inference for the nonconjugate part based on other approximate Bayesian methods.

In future research, three aspects may be of our interest. The first one is to consider accelerated degradation model with covariates and its corresponding VB algorithm. Second, fast Bayesian inference for the RGa model with measurement error is a challenge, which is worth further study. Finally and most importantly, nonlinear degradation path is common in practice, and exploration of fast Bayesian inference for nonhomogeneous RGa model will be considered.

APPENDIX

In this appendix, the observed Fisher information matrix for the proposed model is calculated. With the log likelihood function (36), the first derivatives of $\ell(\Theta)$ with respect to each parameter in Θ are

$$\begin{aligned} \frac{\partial \ell(\Theta)}{\partial \alpha} &= -NMh_c(\psi(\alpha h_c) + \log(\alpha) + 1) \\ &\quad + \sum_{n=1}^N \sum_{m=1}^M h_c \log(x_{n,m}) \\ &\quad + \sum_{n=1}^N A_n(\alpha) \\ \frac{\partial \ell(\Theta)}{\partial \theta_d} &= \sum_{n=1}^N A_n(\theta_d) \end{aligned}$$

where $A_n(s) = \frac{\int_{-\infty}^{\infty} \frac{\partial q(z, y_{n,M}, \Theta)}{\partial s} \exp(-z^2) dz}{\int_{-\infty}^{\infty} q(z, y_{n,M}, \Theta) \exp(-z^2) dz}$, $\theta_d \in \{\mu, \sigma^2\}$, and the first derivatives of $q(z, y_{n,M}, \Theta)$ with respect to each element in Θ are given by

$$\frac{\partial q(z, y_{n,M}, \Theta)}{\partial \alpha} = q(z, y_{n,M}, \Theta) \cdot \log(q(z, y_{n,M}, \Theta)) \cdot \alpha^{-1}$$

$$\begin{aligned}\frac{\partial q(z, y_{n,M}, \Theta)}{\partial \mu} &= q(z, y_{n,M}, \Theta) \cdot (-\log(q(z, y_{n,M}, \Theta))) \\ &\quad - \alpha M h_c \left(\sqrt{2\sigma^2} z + \mu + 1 \right) \\ \frac{\partial q(z, y_{n,M}, \Theta)}{\partial \sigma^2} &= \frac{\partial q(z, y_{n,M}, \Theta)}{\partial \mu} \cdot \frac{z}{\sqrt{2\sigma^2}}.\end{aligned}$$

Next, the second derivatives of $\ell(\Theta)$ in terms of each parameter in Θ are

$$\begin{aligned}\frac{\partial \ell(\Theta)}{\partial \alpha^2} &= -NM h_c (h_c \psi'(\alpha h_c) + 1/\alpha) \\ &\quad + \sum_{n=1}^N [B_n(\alpha, \alpha) + A_n(\alpha)^2] \\ \frac{\partial \ell(\Theta)}{\partial \theta_d \partial \theta_s} &= \sum_{n=1}^N [B_n(\theta_d, \theta_s) - A_n(\theta_d) A_n(\theta_s)]\end{aligned}$$

where $B_n(s_1, s_2) = \frac{\int_{-\infty}^{\infty} \frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial s_1 \partial s_2} \exp(-z^2) dz}{\int_{-\infty}^{\infty} q(z, y_{n,M}, \Theta) \exp(-z^2) dz}$, $\theta_d, \theta_s \in \Theta$ and $(\theta_d, \theta_s) \neq (\alpha, \alpha)$, and the second derivatives of $q(z, y_{n,M}, \Theta)$ with respect to each element in Θ are

$$\begin{aligned}\frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial \alpha^2} &= \frac{\partial q(z, y_{n,M}, \Theta)}{\partial \alpha} \cdot \log(q(z, y_{n,M}, \Theta)) \cdot \alpha^{-1} \\ \frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial \alpha \partial \mu} &= \frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial \mu \partial \alpha} = (\log(q(z, y_{n,M}, \Theta))) \\ &\quad + 1) \times \frac{\partial q(z, y_{n,M}, \Theta)}{\partial \mu} \cdot \alpha^{-1} \\ \frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial \alpha \partial \sigma^2} &= \frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial \sigma^2 \partial \alpha} \\ &= \frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial \alpha \partial \mu} \cdot \frac{z}{\sqrt{2\sigma^2}} \\ \frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial \mu^2} &= \frac{\partial q(z, y_{n,M}, \Theta)}{\partial \mu} (-\log(q(z, y_{n,M}, \Theta))) \\ &\quad - \alpha M h_c \left(\sqrt{2\sigma^2} z + \mu + 1 \right) - 1) \\ &\quad - q(z, y_{n,M}, \Theta) \cdot \alpha M h_c \\ \frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial \mu \partial \sigma^2} &= \frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial \sigma^2 \partial \mu} \\ &= \frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial \mu^2} \cdot \frac{z}{\sqrt{2\sigma^2}} \\ \frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial (\sigma^2)^2} &= \left(\frac{\partial^2 q(z, y_{n,M}, \Theta)}{\partial \mu \partial \sigma^2} - \frac{\partial q(z, y_{n,M}, \Theta)}{\partial \mu} \right. \\ &\quad \left. \times \frac{1}{2\sigma^2} \right) \times \frac{z}{\sqrt{2\sigma^2}}.\end{aligned}$$

Above derivation procedure implies that numerically evaluating (27) is not difficult. First, all integrands evolved in (27) have the structure of GH quadrature. Second, the terms of the second partial derivatives in (27) can be converted to the

simple arithmetic operations in terms of (28), which can be also computed by GH quadrature. Note that evaluating (27) involves only the inverse of a low-dimensional 3×3 matrix, which can be conducted by a number of well-established integrated algorithms, such as the *solve* function in R software. In practice, errors in the inverse calculation of matrices due to computational accuracy overflow are handled by setting the covariance between parameters to a smaller value. This error-handling mechanism allows for more robust operations.

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